

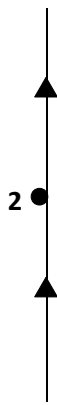
1(a)(i)	$f(0) = 5 + 3 \int_0^0 f(t) dt$ $= 5 - 0$ $= 5$
1(a)(ii)	$f'(x) = 3f(x)$
1(a)(iii)	Let $g(x) = e^{-3x}f(x)$ $g'(x) = -3e^{-3x}f(x) + e^{-3x}f'(x)$ $= e^{-3x}[-3f(x) + f'(x)]$ $= 0 \quad \text{as } f'(x) = 3f(x)$ $\Rightarrow g(x) = a$ for some constant $a \in \mathbb{R}$ From (i), $g(0) = e^{-3(0)}f(0) = 5 \Rightarrow a = 5$, $\Rightarrow g(x) = 5$. Therefore, $5 = e^{-3x}f(x) \Rightarrow f(x) = 5e^{3x}$.
1(b)(i)	By AM-GM Inequality, $1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) \leq \left(\frac{1+3+5+\dots+(2n-1)}{n} \right)^n$ $\leq \left(\frac{\frac{n}{2}(1+2n-1)}{n} \right)^n = \left(\frac{1}{2}(2n) \right)^n = n^n$
1(b)(ii)	$(2n)! = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) \cdot 2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n)$ $\leq n^n \cdot 2^n \cdot n!$ $= (2n)^n \cdot (1 \cdot 2 \cdot 3 \cdot \dots \cdot n)$ $\leq (2n)^n \left(\frac{1+2+3+\dots+n}{n} \right)^n = (2n)^n \left(\frac{\frac{n}{2}(1+n)}{n} \right)^n$ $= (n^2 + n)^n$

2(a)	From MF15, $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ $\Rightarrow \tan A \tan B = \frac{\tan A - \tan B}{\tan(A - B)} - 1$ $\begin{aligned} & \tan 2x \tan x + \tan 3x \tan 2x + \dots + \tan(N+1)x \tan Nx \\ &= \frac{\tan 2x - \tan x}{\tan x} - 1 \\ &+ \frac{\tan 3x - \tan 2x}{\tan x} - 1 \\ &+ \\ &\vdots \\ &+ \frac{\tan(N+1)x - \tan Nx}{\tan x} - 1 \\ &= \frac{1}{\tan x} \left[\begin{array}{l} \tan 2x - \tan x \\ + \tan 3x - \tan 2x \\ + \dots \\ + \tan(N+1)x - \tan Nx \end{array} \right] - N \\ &= \frac{\tan(N+1)x - \tan x}{\tan x} - N \\ &= \tan(N+1)x \cot x - (N+1) \end{aligned}$
2(b)(i)	From the question, $a_{2n-1} = \frac{2n-1+1}{2} = n, \quad a_{2n} = 2^{\frac{2n}{2}}$ Therefore, $b_n = \frac{n}{2^n}$ Now, $S_n = b_1 + b_2 + \dots + b_n = 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^3 + \dots + n\left(\frac{1}{2}\right)^n$ $\Rightarrow \frac{1}{2}S_n = 1\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right)^3 + 3\left(\frac{1}{2}\right)^4 + \dots + n\left(\frac{1}{2}\right)^{n+1}$ $S_n - \frac{1}{2}S_n = \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \dots + \left(\frac{1}{2}\right)^n - n\left(\frac{1}{2}\right)^{n+1}$ $\Rightarrow \frac{1}{2}S_n = \frac{\frac{1}{2}\left[1 - \left(\frac{1}{2}\right)^n\right]}{1 - \frac{1}{2}} - n\left(\frac{1}{2}\right)^{n+1}$

	$= 1 - \left(\frac{1}{2}\right)^n - n\left(\frac{1}{2}\right)^{n+1}$ $= 1 - \left(\frac{n+2}{2}\right)\left(\frac{1}{2}\right)^n$ $\Rightarrow S_n = 2 - (n+2)\left(\frac{1}{2}\right)^n \quad (\text{Shown})$
2(b)(ii)	From (i), since $S_n = 2 - (n+2)\left(\frac{1}{2}\right)^n$ $\Rightarrow 2 - S_n = (n+2)\left(\frac{1}{2}\right)^n$ Let P_n be “$2 - S_n < \frac{1}{n}$ for all $n \geq 6, n \in \mathbb{N}^+$”. When $n = 6$, $2 - S_6 = (6+2)\left(\frac{1}{2}\right)^6 = \frac{1}{8} < \frac{1}{6}$. P_6 is true. Assume P_k is true for some $k, k \in \mathbb{N}$ and $k \geq 6$. i.e. $2 - S_k = (k+2)\left(\frac{1}{2}\right)^k < \frac{1}{k}$ To show P_{k+1} is true, i.e. $(k+3)\left(\frac{1}{2}\right)^{k+1} < \frac{1}{k+1}$. Consider P_{k+1}, $2 - S_{k+1} = (k+3)\left(\frac{1}{2}\right)^{k+1}$ $< (k+3)\left(\frac{1}{2}\right)^k \left(\frac{1}{k(k+2)}\right) \quad \text{from } P_k$ $= \frac{k+3}{2k^2+4k}$ $= \frac{k+3}{k^2+4k+k^2}$ $< \frac{k+3}{k^2+4k+3}$ $= \frac{k+3}{(k+1)(k+3)} = \frac{1}{k+1}$ Therefore P_{k+1} is true. Since P_6 is true and $P_k \Rightarrow P_{k+1}$ is true, P_n is true by mathematical induction.

3(i)	Let $F(x) = f(x) - x$. Notice that F is continuous as well since a difference of continuous functions is continuous. Since $f(x) \in [a, b]$, we have for all $x \in [a, b]$, $a \leq f(x) \leq b \Rightarrow f(x) - a \geq 0 \text{ and } f(x) - b \leq 0.$ In particular, we have $F(a) \geq 0$ and $F(b) \leq 0$. If either $F(a) \geq 0$ or $F(b) \leq 0$ achieves equality, then we are done. Otherwise, we have $F(a) > 0$ and $F(b) < 0$, i.e., $F(b) < 0 < F(a)$. By the intermediate value theorem, there exists $c \in (a, b)$ such that $F(c) = 0 \Rightarrow f(c) - c = 0 \Rightarrow f(c) = c$.
3(ii)	Let $\alpha, \beta \in [a, b]$ be a pair of roots of $f(x) = x$. From the given inequality, we have $\alpha - \beta = f(\alpha) - f(\beta) \leq \lambda \alpha - \beta$, which simplifies to $\alpha - \beta (1 - \lambda) \leq 0$. Since $(1 - \lambda) > 0$, we have $\alpha - \beta \leq 0$. Hence we have, by the Squeeze Theorem, $ \alpha - \beta \leq 0$ $\Leftrightarrow 0 \leq \alpha - \beta \leq 0$ $\Leftrightarrow \alpha = \beta.$
3(iii)	From part (ii), $\begin{aligned} x_n - \alpha &= f(x_{n-1}) - f(\alpha) \\ &\leq \lambda x_{n-1} - \alpha \\ &= \lambda f(x_{n-2}) - f(\alpha) \\ &\leq \lambda^2 x_{n-2} - \alpha \\ &\leq \lambda^3 x_{n-3} - \alpha \\ &\vdots \\ &\leq \lambda^{n-1} x_1 - \alpha \\ &= \lambda^{n-1} a - \alpha . \end{aligned}$ Hence we have $-\lambda^{n-1} a - \alpha \leq x_n - \alpha \leq \lambda^{n-1} a - \alpha $ $\alpha - \lambda^{n-1} a - \alpha \leq x_n \leq \alpha + \lambda^{n-1} a - \alpha $ By limiting n towards infinity throughout the inequality, we have by the Squeeze Theorem, $\lim_{n \rightarrow \infty} [\alpha - \lambda^{n-1} a - \alpha] \leq \lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} [\alpha + \lambda^{n-1} a - \alpha]$ $\Rightarrow \alpha \leq \lim_{n \rightarrow \infty} x_n \leq \alpha$ $\Rightarrow \lim_{n \rightarrow \infty} x_n = \alpha \text{ since } \lambda < 1 \Rightarrow \lim_{n \rightarrow \infty} \lambda^{n-1} a - \alpha = 0.$

4(i)	$\frac{dy}{dx} = \frac{1}{(2x^2 + x - 1)^2} - \frac{(10x + 4)y}{2x^2 + x - 1}$ Euler method: $y(1.5) = y(1) + 0.5 \times \frac{dy(1)}{dx} = 0 + 0.5 \times \left(\frac{1}{4} - 0 \right) = \frac{1}{8}$ $y(2) = y(1.5) + 0.5 \times \frac{dy(1.5)}{dx} = \frac{1}{8} + 0.5 \times \left(\frac{1}{25} - \frac{2.375}{5} \right) = -0.0925$ $y(2.5) = y(2) + 0.5 \times \frac{dy(2)}{dx} = -0.0925 + 0.5 \times \frac{dy(2)}{dx} = 0.037$ $y(3) = y(2.5) + 0.5 \times \frac{dy(2.5)}{dx} = 0.0012295$
4(ii)	$\frac{d}{dx} \left[\ln \left\{ (x+1)^2 (2x-1)^3 \right\} \right]$ $= \frac{d}{dx} \left[\{ 2 \ln(x+1) + 3 \ln(2x-1) \} \right]$ $= \frac{2}{x+1} + \frac{6}{2x-1}$ $= \frac{10x+4}{(x+1)(2x-1)}$ Integrating Factor $= e^{\int \frac{10x+4}{(x+1)(2x-1)} dx}$ $= e^{\int \frac{10x+4}{2x^2+x-1} dx}$ $= e^{\ln \left\{ (x+1)^2 (2x-1)^3 \right\}}$ $= (x+1)^2 (2x-1)^3$ Hence, $(x+1)^2 (2x-1)^3 y = \int \frac{(x+1)^2 (2x-1)^3}{(2x^2+x-1)^2} dx$ $\Rightarrow (x+1)^2 (2x-1)^3 y = \int (2x-1) dx = x^2 - x + C$ $y = 0$ when $x = 1$, $C = 0$ $(x+1)^2 (2x-1)^3 y = x^2 - x$

4(iii)	When $x = 3$ $(3+1)^2 (2(3)-1)^3 y = (3)^2 - 3$ $y = \frac{6}{2000}$ $\% \text{ error for part (i)} = \frac{\left \frac{6}{2000} - (0.0012295) \right }{\left \frac{6}{2000} \right } = 59$
4(iv)	Use smaller step size or Use improved Euler Method.
5(a)	$\frac{dx}{dt} = p(\alpha - L)(\beta - L)$  When $\alpha < L(0) < \beta$, $L \rightarrow \alpha$ as $t \rightarrow \infty$.
5(b)(i)	When $\alpha = \beta = 2$, $\frac{dx}{dt} = p(2 - L)^2$  When $L(0) < 2$, $L \rightarrow 2$ as $t \rightarrow \infty$

5(b)(ii)

When $p = 1$ and $\alpha = \beta = 2$

$$\frac{dL}{dt} = p(\alpha - L)^2 = (2 - L)^2$$

$$\int \frac{1}{(2 - L)^2} dL = \int dt$$

$$\Rightarrow \frac{1}{(2 - L)} = t + c$$

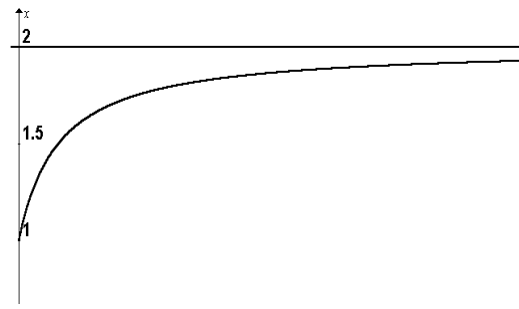
$$\Rightarrow 2 - L = \frac{1}{t + c}$$

$$\therefore L(t) = 2 - \frac{1}{t + c}$$

When $L(0) = 1$, $c = 1$,

the particular solution is $L(t) = 2 - \frac{1}{t + 1}$.

Graph of $L(t) = 2 - \frac{1}{t + 1}$



6(a)

$$\frac{dy}{dt} = 0.06 \left(1 - \frac{y}{12000} \right) y - H$$

$$\text{Let } \frac{dy}{dt} = 0 \quad 0.06 \left(1 - \frac{y}{12000} \right) y - H = 0 \Rightarrow y^2 - 12000y + 200000H = 0$$

$$\therefore y = 6000 \pm \sqrt{6000^2 - 200000H}$$

To sustain population despite harvesting,

$$5500 \geq 6000 \pm \sqrt{6000^2 - 200000H}$$

$$\Rightarrow 0 \leq H \leq 178.875$$

$$\Rightarrow 0 \leq H \leq 178$$

6(b)	To sustain population despite harvesting in the future, $H \leq \frac{kN}{4} = \frac{0.06(12000)}{4} = 180$ Number of squids (in kilotons) to be harvest yearly = 180
7(a)	Auxiliary equation: $r^2 + \lambda r + \mu = 0 \Rightarrow r = \frac{-\lambda \pm \sqrt{\lambda^2 - 4\mu}}{2} = -\frac{\lambda}{2} \pm \frac{i\sqrt{4\mu - \lambda^2}}{2} \quad (\because \lambda^2 < 4\mu)$ General solution: $\theta = e^{-\frac{\lambda}{2}t} \left(A \cos \frac{\sqrt{4\mu - \lambda^2}}{2}t + B \sin \frac{\sqrt{4\mu - \lambda^2}}{2}t \right)$, A and B are arbitrary constant
7(b)(i)	$x = \ln t \quad \frac{dx}{dt} = \frac{1}{t} \quad \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} \Rightarrow \frac{dy}{dt} = \frac{dy}{dx} \frac{1}{t}$ $\Rightarrow t \frac{dy}{dt} = \frac{dy}{dx}$ $\Rightarrow t \frac{d^2y}{dt^2} + \frac{dy}{dt} = \frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dx} \right) \frac{dx}{dt} = \frac{d^2y}{dx^2} \frac{1}{t}$ $\therefore t^2 \frac{d^2y}{dt^2} + t \frac{dy}{dt} = \frac{d^2y}{dx^2}$ Hence $t^2 \frac{d^2y}{dt^2} + t \frac{dy}{dt} - y = 0$ reduces to $\frac{d^2y}{dx^2} - y = 0$
7(b)(ii)	Auxiliary equation: $r^2 - 1 = 0 \Rightarrow r = \pm 1$ General equation: $y = Ae^x + Be^{-x} \Rightarrow y = Ae^{\ln t} + Be^{-\ln t} \Rightarrow y = At + \frac{B}{t}$ When $t = 1, y = 0 \Rightarrow A + B = 0$ ---(1) $y = At + \frac{B}{t} \therefore \frac{dy}{dt} = A - \frac{B}{t^2}$ when $t = 1, \frac{dy}{dt} = 60 \Rightarrow A - B = 60$ ---(2) Solve (1) and (2), $A = 30, B = -30 \therefore y = 30t - \frac{30}{t}$ When $t = 5$, premium paid = $2800 \times 5 = \\$14000$, $y = 30(5) - \frac{30}{5} = 144$ Surrender value \$14400 > \$14000. He would make a profit of \$400.

8(a)(i)	There are 6 solutions as follows: <table><tr><th>Solution</th><th>x_1</th><th>x_2</th><th>x_3</th></tr><tr><td>1</td><td>8</td><td>11</td><td>11</td></tr><tr><td>2</td><td>9</td><td>10</td><td>11</td></tr><tr><td>3</td><td>9</td><td>11</td><td>10</td></tr><tr><td>4</td><td>10</td><td>9</td><td>11</td></tr><tr><td>5</td><td>10</td><td>10</td><td>10</td></tr><tr><td>6</td><td>10</td><td>11</td><td>9</td></tr></table>	Solution	x_1	x_2	x_3	1	8	11	11	2	9	10	11	3	9	11	10	4	10	9	11	5	10	10	10	6	10	11	9
Solution	x_1	x_2	x_3																										
1	8	11	11																										
2	9	10	11																										
3	9	11	10																										
4	10	9	11																										
5	10	10	10																										
6	10	11	9																										
8(a)(ii)	There are 30 identical balls. Let S be the set of all integer solutions. Let $\overline{A_1}$ be the set of integer solutions such that $0 \leq x_1 \leq 10$. Let $\overline{A_2}$ be the set of integer solutions such that $0 \leq x_2 \leq 11$. Let $\overline{A_3}$ be the set of integer solutions such that $0 \leq x_3 \leq 11$. We need $\begin{aligned} \overline{A_1} \cap \overline{A_2} \cap \overline{A_3} &= S - A_1 \cup A_2 \cup A_3 \\ &= S - A_1 - A_2 - A_3 \\ &\quad + A_1 \cap A_2 + A_1 \cap A_3 + A_2 \cap A_3 \\ &\quad - A_1 \cap A_2 \cap A_3 \\ &= \binom{30+2}{2} - \binom{19+2}{2} - \binom{18+2}{2} - \binom{18+2}{2} \\ &\quad + \binom{7+2}{2} + \binom{7+2}{2} + \binom{6+2}{2} - 0 \\ &= 6. \end{aligned}$																												
8(b)	0, since the total capacity of the boxes is less than the supply, i.e., number of balls available.																												
8(c)	The following solution <table><tr><th>Solution</th><th>x_1</th><th>x_2</th><th>x_3</th></tr><tr><td>1</td><td>2</td><td>1</td><td>$n-2$</td></tr></table> is the only one that fails to satisfy the inequality since $x_1 + x_2 + x_3 > n$. Hence there are $3 \times 2 \times (n-1) - 1 = 6n - 7$ solutions.	Solution	x_1	x_2	x_3	1	2	1	$n-2$																				
Solution	x_1	x_2	x_3																										
1	2	1	$n-2$																										

8(d)

Consider the following sums:

$$s_1 = x_1$$

$$s_2 = x_1 + x_2$$

$$s_3 = x_1 + x_2 + x_3$$

$\vdots$

$$s_n = x_1 + x_2 + \dots + x_n$$

If any one of them is divisible by n , then we are done. Otherwise, when each sum is divided by n , the remainder is one of the following $n-1$ possibilities: $1, 2, \dots, n-1$. We create $n-1$ boxes whose names are $1, 2, \dots, n-1$. Then, we place the sums $s_1, \dots, s_n$ into the box labelled with its remainder (when it is divided by n).

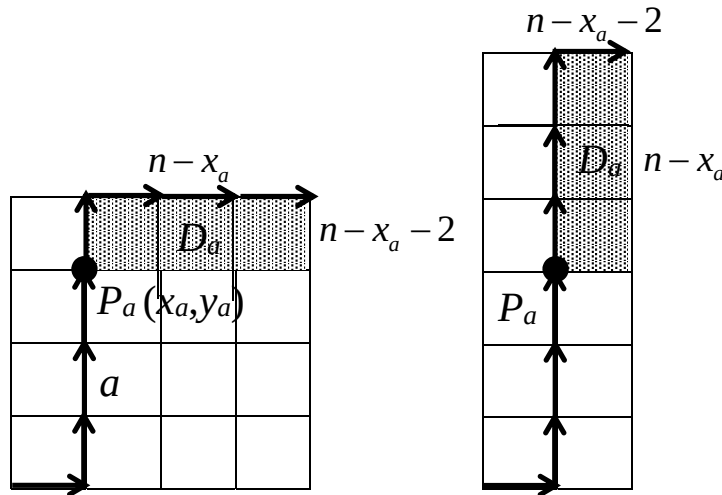
By the Pigeonhole Principle, since $(1)(n-1)+1$ sums are to be placed in $n-1$ boxes, then at least one of them will have $(1)+1$ sums. Let these two sums be s_i and s_k , found in box r , where $i > k$. Now,

$$\begin{aligned} s_i - s_k &= nq_i + r - (nq_k + r) \\ &= n(q_i - q_k). \end{aligned}$$

Hence the required subset is the set of terms of $s_i - s_k$, i.e.,

$$\{x_{k+1}, x_{k+2}, \dots, x_i\}.$$

9(i)

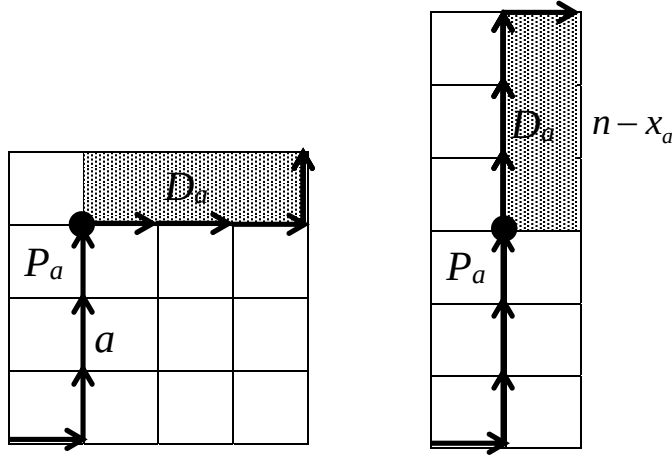


When region D_a is reflected in the line $y = x + 2$, the dimensions of a are swapped.

Hence the new path has dimensions

$$\begin{aligned} (x_a + n - x_a - 2, y_a + n - x_a) &= (x_a + n - x_a - 2, x_a + 2 + n - x_a) \\ &= (n - 2, n + 2). \end{aligned}$$

9(ii)



Let S be the set of paths in the $n \times n$ grid

Let A be the set of paths in the $n \times n$ grid that lie above the line $y = x + 1$ and B be the set of paths in the $(n - 2) \times (n + 2)$ grid obtained when D_a is reflected in the line $y = x + 2$. Let $f : A \rightarrow B$ be the map that reflects the subpath of a that passes through D_a in the line $y = x + 2$. Notice that f is a bijection. Hence by the bijection and complement principles,

$$\begin{aligned}
 |S| - |A| &= |S| - |B| \\
 &= \binom{2n}{n} - \binom{n-2+n+2}{n-2} \\
 &= \binom{2n}{n} - \binom{2n}{n-2} \\
 &= \frac{(2n)!}{n!n!} - \frac{(2n)!}{(n-2)!(n+2)!} \\
 &= (2n)! \left(\frac{(n+1)(n+2)}{n!n!(n+1)(n+2)} - \frac{(n-1)n}{n!n!(n+1)(n+2)} \right) \\
 &= \frac{4n+2}{(n+1)(n+2)} \binom{2n}{n}.
 \end{aligned}$$

10(i)

$$\begin{aligned}
 a_2 &= 1. \\
 a_4 &= a_2 + a_2 + 3 \\
 &= 5.
 \end{aligned}$$

10(ii)

$$a_n = 2a_{\frac{n}{2}} + n - 1.$$

Each half requires a maximum of $a_{\frac{n}{2}}$ comparisons. Following which, there is a maximum number of $n-1$ outputs since we can have the case of alternating outputs from each half. The following example illustrates this alternating concept: For the list $\{3, 1, 4, 2\}$, we have

Step	First Half	Second Half	Output
1.	{3,1}	{4,2}	None
2.	{3}	{1}	None
3.	{ <u>3</u> }	{ <u>1</u> }	{1}
4.	{3}		{1,3}
5.	{4}	{2}	None
6.	{ <u>4</u> }	{ <u>2</u> }	{2}
7.	{4}		{2,4}
8.	{1,3}	{2,4}	None
9.	{ <u>1</u> ,3}	{ <u>2</u> ,4}	{1}
10.	{ <u>3</u> }	{ <u>2</u> ,4}	{1,2}
11.	{ <u>3</u> }	{ <u>4</u> }	{1,2,3}
12.		{4}	{1,2,3,4}